Team 43

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Temperature Detector Robot for Children

Many children tend to cry when they get their temperature checked, most likely because it makes them uncomfortable. We decided to build a device that makes visiting the hospital less dreadful for children who don't enjoy having to be vulnerable during the procedures. We wanted to build a robot that would make temperature-taking enjoyable for children, incorporating elements such as bright colors, music, and a fun-looking robot character. We wanted to ensure that healthcare workers feel less pressure on having to comfort kids and more on ensuring that their main focus is giving the best medical attention. We wanted to create this now because we have not seen much improvement in creating medical devices for simple check-up procedures.

We planned for our robot to lift up one arm after a button is pushed. The arm contains a sensor that reads the temperature, which is then displayed on the LCD screen on the back, as well as an indication of whether the patient is sick or not. A speaker then plays a playful little tune. The device is also easily usable by doctors and parents, with a battery holder on the side that allows for easy replacement of the battery.

For our design criteria, we wanted to focus on cost, sustainability, time/ ease of building, global impact, and fun. We included cost and time/ ease of building because of the constraints we had for this project, such as the limit of $20 for materials and having around 2 weeks to plan and build. The criteria of sustainability were added to follow along with SCU's focus on sustainability. Adding global impact and fun was also crucial to us since we did not want to create a device that did not have a purpose, so we narrowed it down to two designs.

One of the designs was the Temperature Checker Robot which ended up being our final project, which uses a mobile arm to take someone's temperature, then displays it and plays a song. The second was the Sun-Tracking Solar Panel Maximum Energy Collector, which uses 2 light sensors to calculate the angle of maximum sunlight for a solar panel and rotates it accordingly. Our third option was a game that used 2 RGBs, one with a pre-set unique color and

another connected to 3 potentiometers, where the player tries to match the color. The table below shows the three different alternatives in a decision matrix.

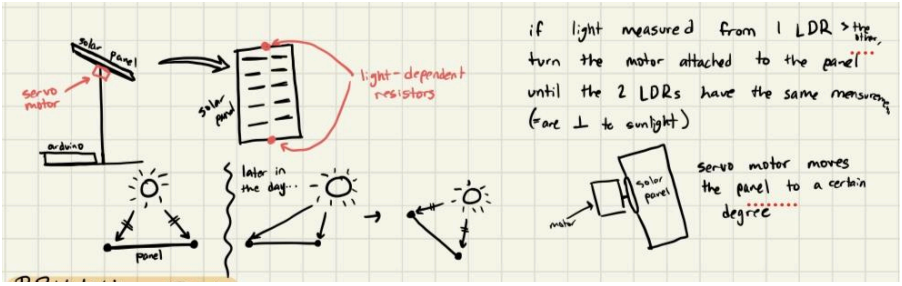


Figure 1: Temperature Sensor Robot Design

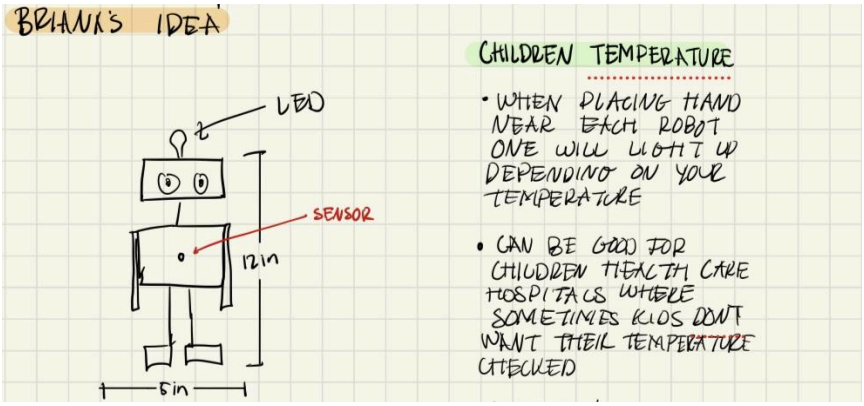


Figure 2: Sun-Tracking Solar Panel Design

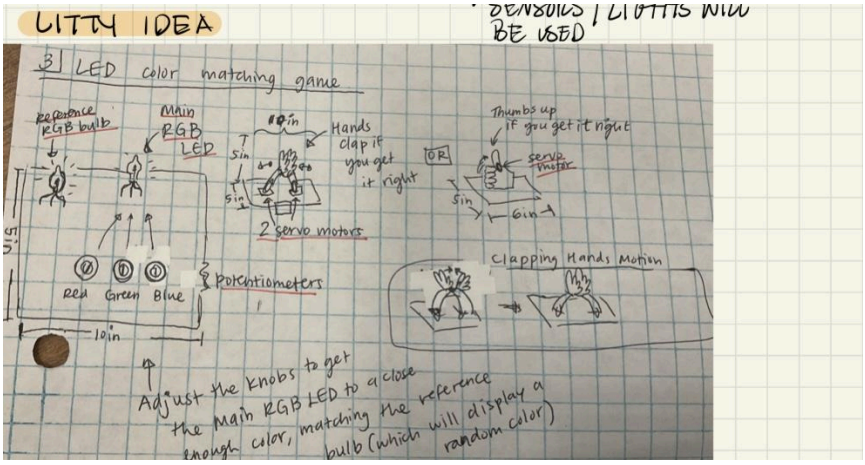
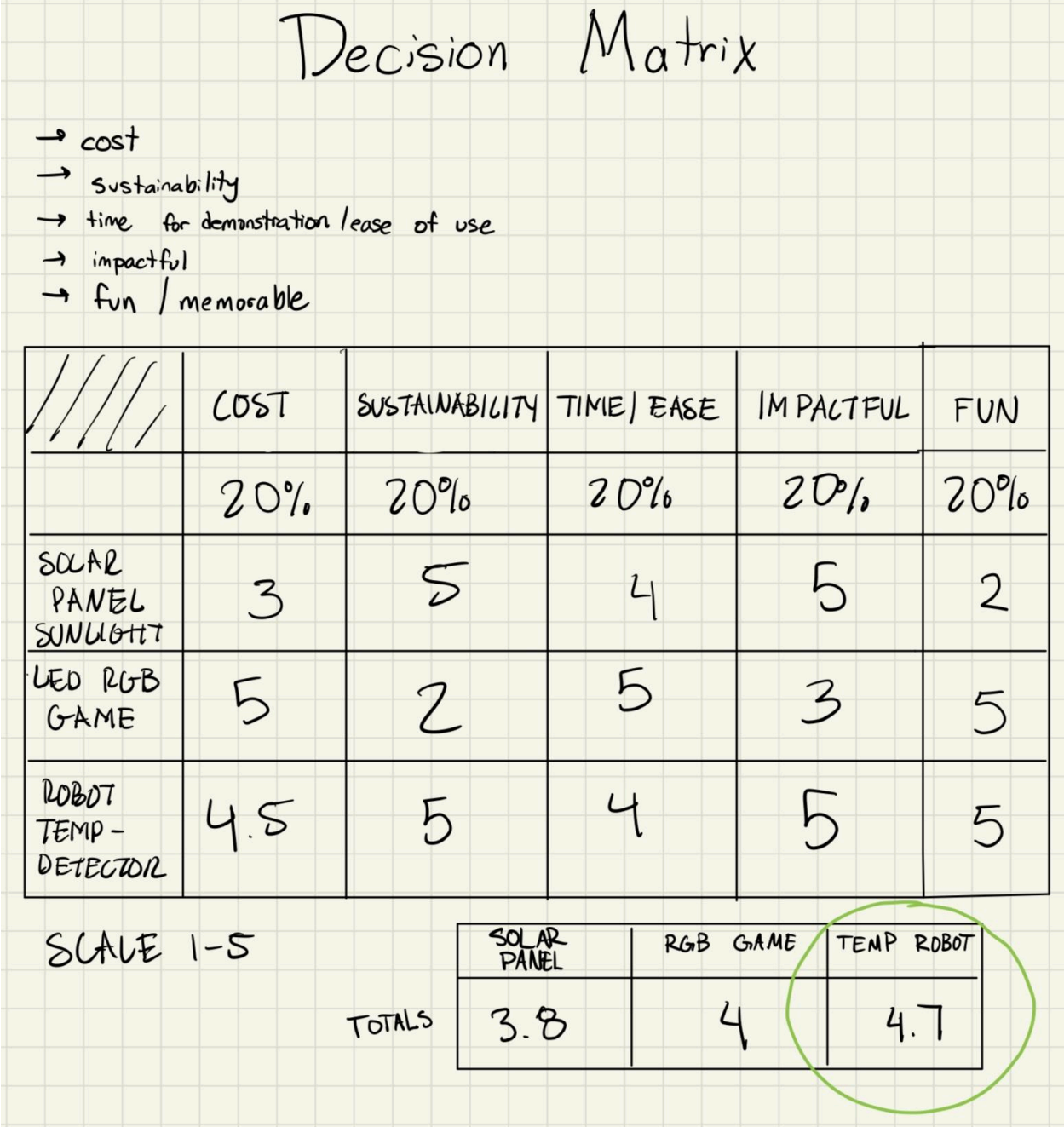


Figure 3: RGB Matching Game Design



////	COST	SUSTAINABILITY	TIME/ EASE	IMPACTFUL	FUN
	20%	20%	20%	20%	20%
SOLAR PANEL SUNLIGHT	3	5	4	5	2
LED RGB GAME	5	2	5	3	5
ROBOT TEMP - DETECTOR	4.5	5	4	5	5

	SOLAR PANEL	RGB GAME	TEMP ROBOT
TOTALS	3.8	4	4.7

Figure 4: Decision Matrix

Final design details:

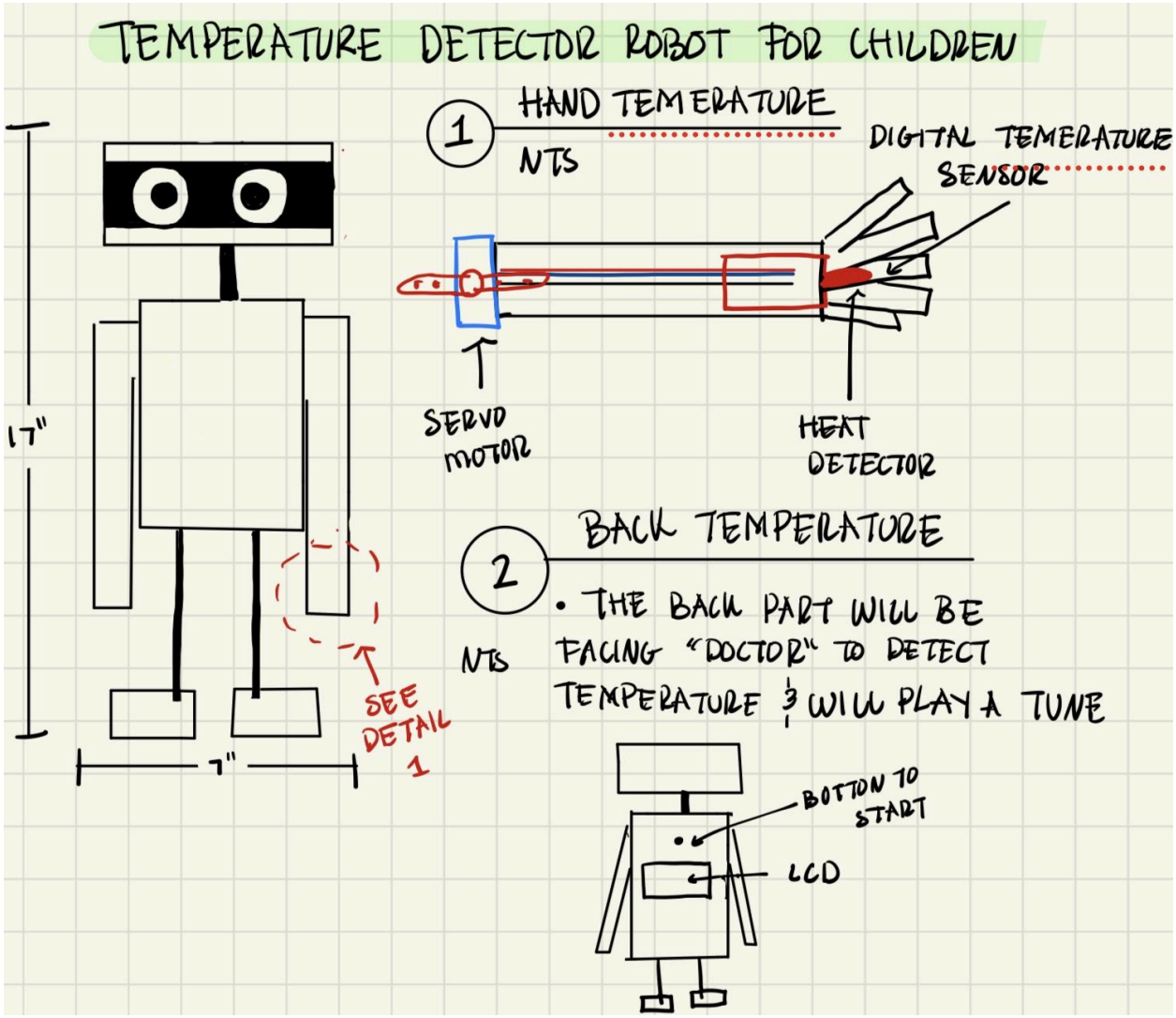


Figure 5: Final Design Chosen: Temperature Detector Robot for Children

After deciding on the top three choices from all 9 project ideas, we decided on the Temperature Detector Robot for Children. A key feature is when we press on the push button that is on the back of the robot it moves the hand up which is essentially getting close to the child patient. The child touches its hand and it will determine the temperature. On the back of the robot, there is an LCD screen that says “Hello World” to begin and then displays the temperature. If the temperature is too high, the display recognizes this and says “Sick”. Otherwise, the display tells the doctor or parent “healthy!” to indicate the child has a normal body temperature. Finally, it

plays a happy tune to cheer up the child. Our robot was created at a scale that is easily mobile so in a hospital setting, the healthcare worker can move it to different locations. It is 7 inches by 17 inches.

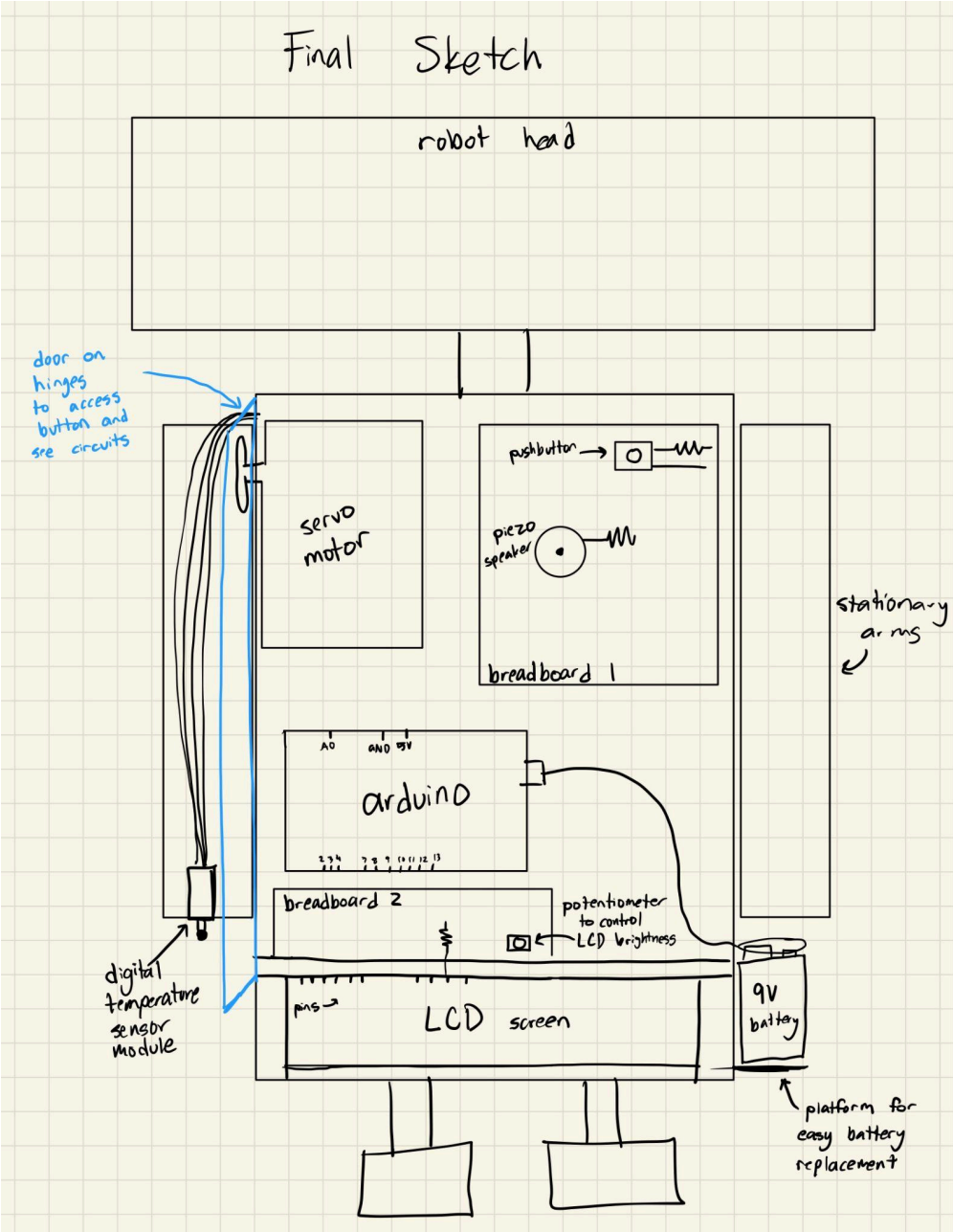


Figure 6: Electronic Schematic

The image above shows the inside of the robot and all of the electronics schematic, including where the LCD screen is located, an Arduino, two breadboards, and a servo motor to move the arm. In the robot's body, we made sure to create enough space so that all the wiring could fit

nicely without impacting the robot's performance. An advantage to this setup is that we can clearly see the efficiency of the circuits and also easily replace them since the back of the robot can open and close. This ties in with the category of sustainability since it will be easy for a user to replace any of the hardware. Apart from the hardware, the materials used were also recyclable, and one one 9V battery is needed to power the entire thing.

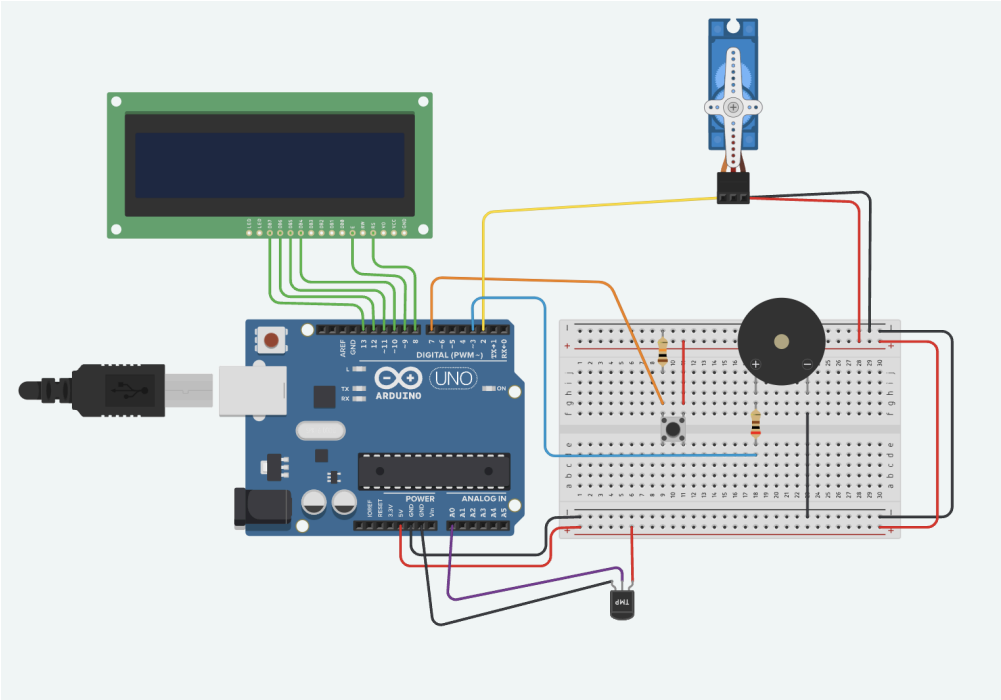


Figure 7: TinkerCad Circuit

Item	Quantity	Description	Cost per item	Total Cost
01	1	Cardboard	$5	$5
02	6	Paint Color Bottles	$2	$12
03	1	Tape	$1.84	$1.84
04	1	Digital Temperature sensor	N/A - ENGR 1 loan	
05	1	Piezo speaker	N/A - ENGR 1 loan	
06	1	Arduino UNO	N/A - ENGR 1 loan	
07	1	Push button	N/A - ENGR 1 loan	
08	1	10 kΩ resistor	N/A - ENGR 1 loan	
09	1	200 Ω resistor	N/A - ENGR 1 loan	
10	1	Servo motor	N/A - ENGR 1 loan	
11	2	Breadboard	N/A - ENGR 1 loan	
		Sum of all parts - Total Cost		$18.84

Table A1: Bill of Materials for Temperature Detector Robot project

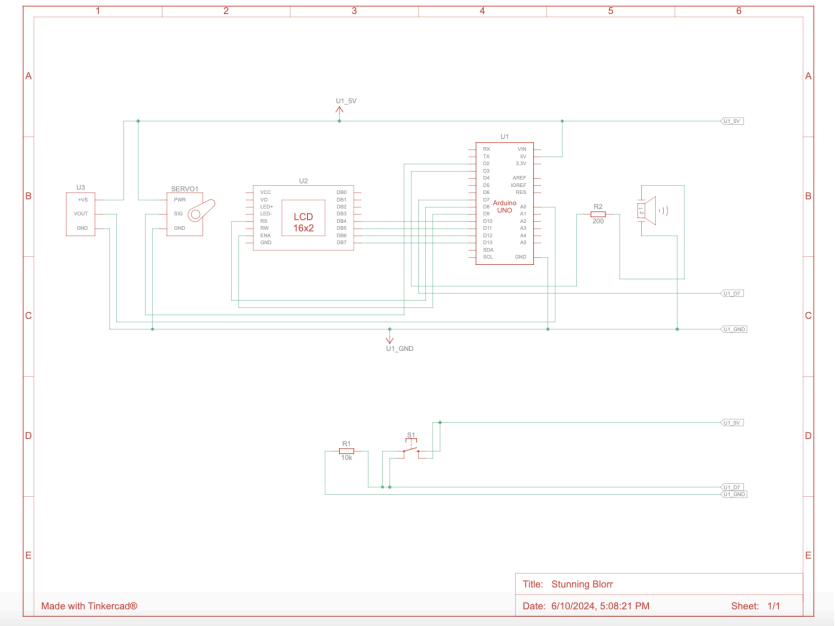


Figure A2: Electronic schematic diagram for robot project

Team Member	Time spent on project (include individual and team time)	
	In-class (any lab periods attended)	Out-of-class (non-lab periods)
Kate Picca	6 hrs	12 hrs
Briana Renteria Lopez	6 hrs	12 hrs
Litty Lalu	6 hrs	12 hrs

Table A3: Team member project time log

Commented computer code for robot project:

```
//include the appropriate libraries
#include <Servo.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include <Math.h>
#include "pitches.h"
#include <LiquidCrystal.h>
//liquid crystal display pin numbers:
LiquidCrystal lcd(8, 9, 10, 11, 12, 13);
const int buttonPin = 7;
const int speakerPin = 3;
byte NTCPin = A0;
//define melody with pitches and note durations in arrays
int melody[] = {NOTE_C4, NOTE_G3, NOTE_G3, NOTE_A3, NOTE_G3, 0, NOTE_B3, NOTE_C4};
// note durations: 4 = quarter note, 8 = eighth note, etc.:
int noteDurations[] = {4, 8, 8, 4, 4, 4, 4, 4};
Servo myservo;//arm servo
int buttonState = 0;
int servoPos = 225;//initial servo position
int pos;
//temperature values:
float testVal = 3.0;
const float R1 = 10000;//value of fixed resistor in voltage divider circuit
const float logR2 = 7.7613;//ln(resistance of thermistor at reference temp - 25C)
const float T1 = 298.15;//reference temp in Kelvin
const float B = 3950;//beta coefficient of thermistor
float analogToCelsius(int rawADC) {
```

```
float Vout = (rawADC / 1023.0) * 5.0; // Convert ADC reading to voltage
float R2 = R1 * (5.0 / Vout - 1.0); // Convert voltage to resistance
float logR2_R1 = log(R2 / R1);
float temp = 1.0 / (1.0 / T1 + logR2_R1 / B);
temp -= 273.15; // Convert temperature to Celsius
return (temp);}
void setup()
{//set up sensors
Serial.begin(9600);
pinMode(buttonPin, INPUT);
pinMode(speakerPin, OUTPUT);
myservo.attach(2);//myservo on pin 2
myservo.write(servoPos);//initial position
lcd.begin(16, 2);
lcd.print("hello world!");
delay(2000);}
void loop()
{buttonState = digitalRead(buttonPin);
if(buttonState == HIGH){//if button pressed
  myservo.write(servoPos);//initial position
  for (pos = servoPos; pos > 90; pos --) { // moves arm from 0 to 90 deg
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);}
  delay(3000);//wait 3 sec for doctor to place robot on forehead
  //measure temp and display
  int rawADC = analogRead(A0); // Read analog value from the thermistor
  float temperatureC = analogToCelsius(rawADC); // Convert analog reading to Celsius
  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" °C");
  delay(1000); // Delay for readability, adjust as needed
  lcd.clear();
  delay(1000);
  lcd.setCursor(0, 0);
  lcd.print("Temperature:");
  lcd.print(temperatureC);
  lcd.print(" °C");
  delay(2000);
  //if temp is hot (fever temp) we chose a lower temp to test for to show that the robot can sense both sick and
```

```
healthy temps without needing a heating device or sick person
  //this number can be adjusted in the if statement below
  if(temperatureC >= 28.0){
    lcd.clear();
    lcd.print("sick :("); }
 //if normal temp
  else{
    lcd.clear();
    lcd.print("healthy!");}
  delay(1000);
  for (pos; pos < 225; pos ++) { //move arm back to initial position
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15ms for the servo to reach the position}
  delay(1000);
  //play tune:
  for (int thisNote = 0; thisNote < 8; thisNote++) {
    int noteDuration = 1000 / noteDurations[thisNote];//to calculate the note duration: one second(1000ms)
divided by the note type
    tone(speakerPin, melody[thisNote], noteDuration);//play each pitch
    int pauseBetweenNotes = noteDuration * 1.30;
    delay(pauseBetweenNotes);//delay for 30% longer than noteDuration for each note
    noTone(speakerPin);//stop the tune playing}
  delay(1000);
  lcd.clear(); }}
```

Figure 8: Final Commented Code

Conclusion:

Overall this robot can serve an important purpose in any hospital since it is cost-efficient, sustainable, impactful, and fun. These traits are important to us since we as engineers are aware of the impact and our duty to serve our community. In this design, we are targeting children who can be inspired by the devices created in our generation to later continue to create devices that can also impact the following generation. Our Temperature Detector for Children can become essential to a medical hospital and make the medical field more fun and less stressful for many young children.